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A Summary and Critique of the research paper 'Kefir drink leads to a similar weight loss, compared with milk, in a dairy-rich non-energy-restricted diet in overweight or obese premenopausal women: a randomized controlled trial'.

Introduction-

Obesity is a major healthcare concern around the world, trials are being conducted using dairy products due to their chemistry that benefits energy balance and aids weight loss, but there are currently mixed opinions on whether there is significant evidence that increasing dairy without energy restriction can lead to weight loss.^[1] I will be critiquing and summarising a research paper on the first RCT (randomised control trial) study conducted in Iran to compare the potential weight-reducing effects of kefir drink and milk in a dairy-rich non-energy-restricted diet in overweight and obese premenopausal women.^[2]

Summary-

This study is designed as a single-centre, multi-arm, parallel-group RCT lasting 8 weeks. The primary objective set by the researchers was to measure the change in weight. The secondary objectives were to investigate changes in BMI and WC (waist-circumference)- all objectives are well-defined throughout the report. The researcher's hypothesis has been explored previously with research into the weight-reducing effects by dairy products having been conducted however this is the first RCT comparing these exact products and dosage in women. Eligible participants were recruited from outpatients referred to the Cardiovascular Research Centre in Iran and were required to be 25-45 years old, have a BMI of 25.0-34.9 kg/m² and have regular menstruation cycles. Results from this test were that at 8 weeks, there were significant decreases in weight, BMI and WC ($p < 0.001$) in the Kefir and milk groups compared to the control group.

Technical Details:

Study Set Up- The sample size needed for this study was determined using standard deviations in previous similar studies, so a sensible number of participants was used. Simple randomization was used to split participants into 1 of 3 groups- kefir, milk and control. The random allocation used ensures that the groups have identical distributions of characteristics and removes bias.

The study clinician assessing eligibility was blinded to treatment allocation however, the participants were not. Therefore, the gold standard, double blind trial was not used- this could introduce bias into the data, especially in the control group as this could explain why weight loss was observed amongst all 3 groups.

Study Analysis- There were some dropouts in this study, so all analyses conducted used both intention-to-treat (including all participants, even 'dropouts') and per-protocol (only using participants that followed study protocol) analysis- no significant results were found in one without being significant in the other. A total of 75 participants were eligible but 58 participants were included in the per-protocol analysis due to other participants leaving the trial.

The continuous variables assessed in this RCT were checked for normality using Kolmogorov-Smirnov test (where a high p-value in the test would indicate variables following the normal distribution)^[3] and either a one-way analysis of variance or Kruskal-Wallis H test to compare for normality in variables amongst the 3 groups. The Kruskal-Wallis H test was used since we have 3 different groups and is appropriate to use as the variable of interest is continuous, there is

enough data, and there is a similar spread found amongst all groups. It also can be used only if the groups were assigned by random sample, like in this trial.^[4]

Analysis between the groups was conducted pairwise by a Bonferroni post hoc test or Mann-Whitney U test. The post hoc test is used as we know of there being a different mean between the groups, but we want to determine which of the groups' means are different and also, it can adjust for multiple comparisons between our 3 groups.^[5] A Mann-Whitney U test can be used to test whether 2 different groups have the same shaped data but requires the continuous variables to not have a normal distribution which was assessed using the Kolmogorov-Smirnov test but results from this were not clear in the report.^[6]

After the differences were calculated at 8 weeks, the change in weight, BMI and WC were compared using analysis of covariance (ANCOVA) which allowed the comparison between the 3 groups to be made whilst controlling for at least one continuous covariate which could influence the primary objectives, i.e. controlling for baseline dietary calcium intake.^[7]

Study Results- Results were reported as means and 95% confidence intervals alongside the p-value of appropriate test. No other statistics have been analysed and there is a lack of plots to present their findings. Looking at Table 1, we can clearly see there is statistical difference found between the dairy products and control and none between Kefir and milk in Weight and BMI in the Intention-to-Treat data. The results were concluded using pairwise comparisons- subjects in the kefir and milk groups had significantly greater reductions in weight and BMI after 8 weeks compared to those in the control group ($p < 0.01$) but no significant differences were found in anthropometric indices. The unadjusted pairwise differences are not statistically significant between Kefir-control but are statically significant in Milk-control so there could be some valuable information there which isn't reported on in the paper. However, the adjusted values are significant.

ITT population (n = 75)	Kefir (n = 25)	Milk (n = 25)	Control (n = 25)	p value ^a	Unadjusted pairwise between group difference ^b					
					Kefir-milk	p value	Kefir-control	p value	Milk-control	p value
Unadjusted values										
Weight (kg)	-2.2 [-2.5; -1.9]	-2.1 [-2.2; -1.9]	-1.2 [-1.4; -0.9]	<0.001	-0.2 [-0.6; 0.3]	1.000	-1.1 [-1.5; -0.6]	<0.001	-0.9 [-1.4; -0.4]	<0.001
BMI (kg/m ²)	-0.9 [-1.0; -0.8]	-0.8 [-0.8; -0.7]	-0.4 [-0.6; -0.3]	<0.001	-0.1 [-0.3; 0.1]	0.504	-0.4 [-0.6; -0.3]	<0.001	-0.3 [-0.5; -0.2]	<0.001
WC (cm)	-2.1 [-2.5; -1.6]	-1.9 [-2.4; -1.3]	-1.1 [-1.4; -0.7]	0.004	-0.2 [-0.9; 0.6]	1.000	-1.0 [-1.8; -0.2]	0.005	-0.8 [-1.6; -0.1]	0.028
Adjusted pairwise between group difference ^b										
					Kefir-milk	p value	Kefir-control	p value	Milk-control	p value
Adjusted values										
Weight (kg)	-2.2 [-2.5; -2.0]	-2.1 [-2.3; -1.8]	-1.1 [-1.4; -0.9]	<0.001	-0.2 [-0.6; 0.3]	1.000	-1.1 [-1.5; -0.7]	<0.001	-0.9 [-1.3; -0.5]	<0.001
BMI (kg/m ²)	-0.9 [-1.0; -0.8]	-0.8 [-0.9; -0.7]	-0.4 [-0.5; -0.4]	<0.001	-0.1 [-0.2; 0.1]	1.000	-0.4 [-0.6; -0.2]	<0.001	-0.4 [-0.5; -0.2]	<0.001
WC (cm)	-1.9 [-2.3; -1.5]	-2.0 [-2.4; -1.6]	-1.1 [-1.5; -0.7]	0.001	0.1 [-0.6; 0.8]	1.000	-0.8 [-1.5; -0.2]	0.008	-0.9 [-1.5; -0.3]	0.002

Table 1: A screenshot of Table 3 from the research paper.

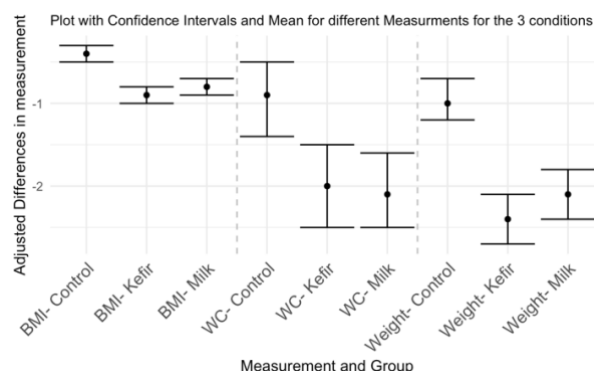
This research is clinically important as it highlights that no significant difference can be found between milk and kefir. Milk is more readily available and therefore more cost effective so would be beneficial for clinicians to be made aware of this research. Since there is a worldwide problem with obesity, having a product that is easily accessible could significantly reduce weight for numerous overweight individuals, helping to solve one of the main healthcare issues in the world today.

Discussion-

The main weakness of this RCT is that we can only conclude results for 8 weeks- in those 8 weeks, there are a plethora of confounding variables that could affect weight loss. The study has tried to minimise the impact of these confounding variables as much as possible but since this study relies on the participants to self-report it comes with a lot of recall bias as we cannot

be certain that they have told the truth or have remembered whether they stuck to study protocol. Furthermore, the lack of double blinding could add the possibility of bias. Also, there are numerous studies which have been conducted with opposing evidence that ran for much longer- for example, comprehensive research was conducted on 29 RCTs with similar variables of interest but are long-term studies without energy-restriction. These found that consumption of dairy products did not result in a significant reduction in weight, but it did in the context of energy restriction and short term.^[8] A longer study would be my advice for any further research.

One of the limitations discussed in the report is that we cannot make inferences about other populations such as men, so the results are not very generalisable. More subjects of a different population pool (men, older, higher/lower BMI etc.) could improve generalisability.



Due to a lack of plots in the original report, I have plotted the adjusted differences from Table 1, which we can see in Figure 1- as some visualisation of the data can provide easier interpretation. For weight- there is some slight variation between Kefir and Milk- although nothing statistically significant, further research and longer study time could let us explore this difference more.

Figure 1- The adjusted differences from baseline results in for the Per-protocol population plotted as confidence

and any differences can be assumed as a causal effect of treatment. In Figures 2 and 3, I have plotted the mean and confidence intervals for the variables of interest from the per-protocol population- we can see that all are similar across the three groups but there are slight differences in the confidence intervals, and we observe that the participants in kefir have a slightly bigger WC and BMI. These differences are small, but a potential improvement could be the use of pairing in the groups, so they are matched similarly with the variables of interest (Weight, BMI and WC).

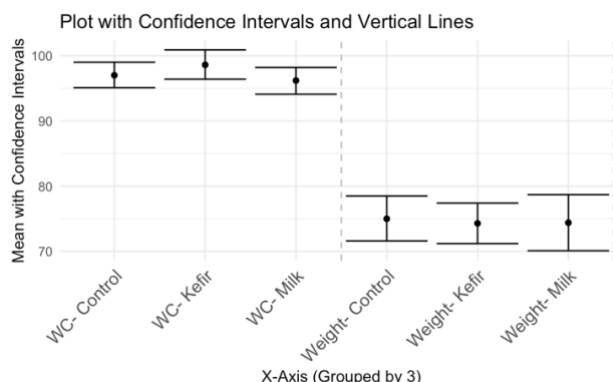


Figure 2- Baseline measurements for WC and Weight, plotted as confidence intervals.

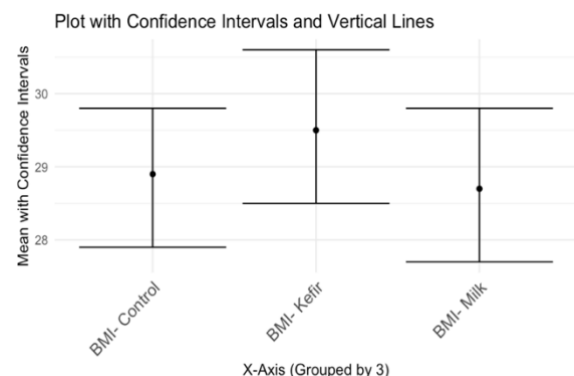


Figure 3- Baseline measurements for BMI, plotted as confidence intervals

The report also looked at other studies that investigated the differences in weight at different dosages of different yoghurts which also ran for 8 weeks- they investigated risk factors and found significant differences at different dosages.^[9] Therefore, another improvement could be an increased amount of groups with different amounts of kefir and milk as we could find out whether there is a linear increase in weight loss. For example, we could also explore if there is any significant difference between 1 and 2 additional servings.

Conclusion-

To conclude, the research that has been conducted in this trial is vital and there is good evidence to suggest that an increased dairy intake without energy restriction will lead to weight loss, as there are significant differences between the kefir and milk groups when compared to the control. However, for this research to be more viable their needs to be a long-term study conducted and a bigger sample size with wider eligibility so the results can have improved generalisability, i.e. including men.

Word Count: 1511 (Including 50 words in captions of Figures and Tables)

A.I Generative Statement- I have used 'Chat GPT' to aid making the plots in R only.

References:

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